



Name: _____

Class: _____

Mr. Rawson • GCSE Computer Science

Year 11 2026

Programming Quiz 1 — MARK SCHEME

Time: 25 minutes Total: 25 marks

Answer **all** the questions. Write your answers on the dashed lines. You may write code in Python or in OCR Exam Reference Language.

Every question here is a **real OCR exam question**, set out the way it appears on the paper. Question 9 is a stretch: do it last.

1 Tick (✓) **one** box in each row to identify the programming construct where each keyword is used.

Keyword	Programming construct	
	Selection	Iteration
if	✓	
for		✓
while		✓

[3]

1 mark per row. *Examiner:* answered very well by most candidates.

OCR J277 · ExamBuilder Q56

2 Write pseudocode to increment a value held in a variable `score` by one.

score = score + 1

----- [1]

Accept `score += 1`, or structured English (“score becomes score plus one”). **Do not accept** `score + 1` or `score = +1`. *Examiner:* most added one; fewer assigned the result back to `score` — that assignment is the mark.

OCR J277 · ExamBuilder Q47(a)

- 3 The result of a computer game can be a win, a loss or a draw. The result is stored in a variable with the identifier `result`.

(i) State why `result` **cannot** be stored using a Boolean data type.

Boolean has only two values (True / False); result has three.

[1]

(ii) Give a suitable data type for `result`. Explain your answer, showing how each result could be stored.

Data type **String**

Explanation **can store "win", "loss" or "draw"**

[2]

(ii) 1 mark type, 1 mark matching explanation. Also accept **Char** ("W", "L", "D") or **Integer** (0, 1, 2).

OCR J277 · ExamBuilder Q51(b)

- 4 In a computer game, a player collects items. Each item has a score. The score for each item is stored using the data type **real**.

State what is meant by the data type **real**.

A number with a decimal / fractional part.

[1]

Do not accept an example on its own (e.g. "3.5").

OCR J277 · ExamBuilder Q51(c)

- 5 Tick (✓) **one** box in each row to identify if each operator is a comparison operator or an arithmetic operator.

Operator	Comparison	Arithmetic
==	✓	
+		✓
DIV		✓
>	✓	

[4]

1 mark per row.

OCR J277 · ExamBuilder Q52

6 DIV and MOD are both operators used in computing-related mathematics.

(i) State the value of `13 DIV 4`

3 [1]

(ii) State the value of `13 MOD 4`

1 [1]

Correct answer only.

OCR J277 · ExamBuilder Q38

7 (a) A programmer declares the following variables.

```
first = "Computer Science"
second = "is great"
```

Strings can be concatenated (joined together) using the `+` operator. For example, `print("Maths" + second)` will output `Maths is great`

Use string manipulation with the variables `first` and/or `second` to produce the following output.

(i) `great`

second.substring(3,5) or second[3:] [1]

(ii) `Computer`

first.substring(0,8) or first[:8] [1]

(iii) `Science is great`

first.substring(9,7) + " " + second or first[9:] + " " + second [1]

(b) State the output from the following lines of program code.

(i) `print(first.length)`

16 [1]

(ii) `print(second.length DIV 3)`

2 [1]

(iii) `print(3 ^ 2)`

9 [1]

(a) Ignore `print` or its absence. Accept any valid string method that uses the variables, in any language. (iii) must produce the space correctly. *Examiner:* candidates were “generally unfamiliar with substring operations”; – cannot remove text from a string, so `"is great" - "is "` scores nothing. (b)(ii): `second` has 8 characters, and `8 DIV 3` is 2.

OCR J277 · ExamBuilder Q41(a)(b)

8 Write an algorithm in pseudocode to:

- store "Hello" in the variable word1
- store "Everyone" in the variable word2
- concatenate word1 and word2 to store "HelloEveryone" in the variable message

word1 = "Hello"

word2 = "Everyone"

message = word1 + word2

[3]

1 mark per bullet. Accept &, +, . or concat(). **Do not allow** a comma, == for assignment, or spaces in variable names (word 1). *Examiner:* the question asks to *store* the result, not print it — printing instead of storing lost the mark.

OCR J277 · ExamBuilder Q55(a)

9 **Stretch.** Louise writes a program to work out if a number entered by the user is odd or even. Her first attempt at this program is shown.

```
01 num = input("enter a number")
02 if num MOD 2 >= 0 then
03     print("even")
04 else
05     pritrn("odd")
06 endif
```

The program contains a logic error on line **02**.

(i) State what is meant by a logic error.

An error that gives incorrect output, but the program still runs.

[1]

(ii) Give a corrected version of line **02** that fixes the logic error.

if num MOD 2 == 0 then

[1]

(i) Do not accept an example of a logic error. *Examiner:* "when code does not work" is not precise enough — the idea is wrong output *without* stopping. (ii) The point is >= becomes ==; accept Python num % 2 == 0, and ignore casting. pritrn on line 05 is a *syntax* error, not the one asked about.

OCR J277 · ExamBuilder Q23

END OF QUIZ · 25 marks